TP3 KERNEL DENSITY ESTIMATION KDE :

- **Obligatory**
- **Individual work**
- **Deadline**: 13/04/2020
- **Penalty** -20% for each day of delay

Single Variable KDE

TRN = $\{x_i\}$, x a new instance

→ Estimate $p(x)$

- Center the kernel K on each x_i
- $\forall$ tr. point x_i : contributes to the $p(x)$ density according to its distance from x
- $$\hat{p}(x) = \frac{1}{n \underbrace{V_R}_{=h}} \sum_{i=1}^{n} K\left(\frac{\overbrace{x - x_i}^{x - x_i}}{h}\right)$$

Gaussian kernels

- Univariate kernel $$K_1(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} x^2\right) \quad (1)$$
- Multivariate kernel $$K_d(x) = \frac{1}{(2\pi)^{d/2}} \exp\left(-\frac{1}{2} x^T x\right) \quad (2)$$

To estimate density function $p(x)$

$$p(x) \simeq \frac{1}{n V_R} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right) \quad (3)$$

d : dimentionality (# attributes)

n : # tr. points

h : edge length

V : Volume

univariate $d=1$

$V = h$

multiv : $V = h^d$

↳ $x \in \mathbb{R}^d$ (multivariate vector)

2 ways to work:

(A) Use univariate version of kernel

· fit 1 kernel to each of the d dimentions

new instance (test) center (training point)

(3) $\Rightarrow p(x) = \frac{1}{nV_R} \sum_{i=1}^{n} \prod_{j=1}^{j=d} K\left(\frac{x_j - x_{ij}}{h}\right)$ (product of univariate kernels)

$$p(x) \overset{(1)}{=} \frac{1}{nV_R} \sum_{i=1}^{n} \prod_{j=1}^{d} \frac{1}{(2\pi)^{1/2}} \exp\left(-\frac{1}{2} \frac{(x_j - x_{ij})^2}{h}\right)$$

$$= \frac{1}{nV_R} \sum_{i=1}^{n} \frac{1}{(2\pi)^{1/2}} \exp\left(-\frac{1}{2}\left(\frac{x - x_i}{h}\right)^2\right)$$

B) Use multivariate version of kernel

$$3 \overset{(2)}{\Rightarrow} p(x) = \frac{1}{nV_R} \sum_{i}^{n} \frac{1}{(2\pi)^{d/2}} \exp\left(-\frac{1}{2} \frac{(x - x_i)^T}{h} \frac{(x - x_i)^T}{h}\right)$$

$$= \frac{1}{n} \sum_{i} \frac{1}{(2\pi)^{d/2} |H|^{1/2}} \exp\left(-\frac{1}{2} (x - x_i)^T H^{-1} (x - x_i)\right)$$

H : diag matrix with all its diag equal to h^2.

$|H| = h^{2d} = V_R^2$